Andrew Detroyer

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Professional Summary

Analytical and solutions-driven Software Engineer with several years of programming experience, specialized in full-stack development, advanced AI integrations, and infrastructure automation. Proven capability in building bespoke technical solutions, ranging from training machine learning models on custom data to orchestrating agentic automation pipelines and engineering enterprise-grade local infrastructure.

Technical Skills

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  Languages & Frameworks: Swift, SwiftUI, SwiftData, Python, Node.js, Bash, HTML/CSS
- **AI & Machine Learning:** Custom Model Training, Dataset Curation, NVIDIA NIM, Agentic Workflows (Claude, Gemini)
- **DevOps & Infrastructure:** Proxmox VE, LXC Containers, Debian, Docker, Home Assistant, UniFi Networking, Dell PowerEdge Server Management
- **Systems & Tools:** Git/GitHub, Xcode, Synology Storage, API Integration (Google Places API, REST)

Projects

Custom Machine Learning Model & Data Curation

- Engineered and trained a proprietary AI model published to GitHub, leveraging self-curated datasets to solve domain-specific problems.
- Structured data preprocessing pipelines to optimize training efficiency and model accuracy.
- Integrated intelligent workflows into localized computing environments, enabling automation scripts to interact seamlessly with AI agents.

Personal Calorie & Macro Tracker Application

- Architected and developed a full-stack iOS application from scratch using Xcode, SwiftUI, and SwiftData for localized persistence.
- Implemented an advanced photo-to-macro parsing feature utilizing NVIDIA NIM microservices to extract precise nutritional data from imagery.
- Tailored application logic to support strict dietary tracking configurations, handling micro/macro inputs dynamically.

Media Infrastructure Optimization & Automation

- Formulated a migration and deployment strategy moving high-volume media services to a dedicated Debian 12 server hosted on enterprise-grade Dell PowerEdge R630 hardware.

- Developed custom automation scripts to coordinate background tasks, data transfer, and library syncs between local compute nodes and a Synology NAS array.
- Established containerized microservices via Proxmox LXC to compartmentalize computing workloads and maximize server uptime.

Lead Generation & Data Extraction Engine

- Developed a customized automation script using the Google Places API to identify regional small businesses lacking a web presence.
- Parsed complex JSON responses to extract high-value B2B sales targets across major commercial areas.

Experience

Independent Software Developer / Systems Engineer *2022 – Present*

- Designed, tested, and maintained open-source repositories and personal production systems.
- Flashed, modded, and configured custom hardware firmware (including wireless routers and electric vehicle motor controllers) to unlock performance optimizations.
- Architected local network telemetry systems utilizing ESPHome and ESP32-C6 hardware for environment tracking and data logging.

Education

Coursework & Certifications in Computer Science & Applied Sciences

- **Honors Recognition:** Awarded Honors Cord for academic excellence and top-tier performance in Advanced Programming.
- **Core Focus Areas:** Systems Design, Algorithmic Logic, Advanced Mathematics, and Economics.